When Sample Selection Bias Precipitates Model Collapse

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Introduction

Synthetic data now flows back into future training sets, so recursive training can amplify small distributional errors into model collapse (Shumailov et al., Nature 2024). Verification and sample selection are meant to stop this loop by keeping samples that match a trusted reference. In low-resource communities, however, that reference is local and incomplete: valid rare modes can look like bad samples and be filtered away.

Theory

Proposition 1: Replace paradigm (Shumailov et al., Nature 2024)

Under the *Replace* paradigm, as $t \to \infty$,

$$\Sigma_t \xrightarrow{a.s.} \mathbf{0}, \qquad \mathbb{E}[\mathbb{W}_2^2(\mathcal{N}_t, \mathcal{N}_0)] \to \infty.$$

Replacing data eventually removes variance and drifts away.

Proposition 2: Accumulate paradigm (Kazdan et al., ICML 2025)

Under unbiased accumulation,

$$\mathbb{E}[\bar{\Sigma}_t] \to \alpha_n \bar{\Sigma}_0.$$

Thus $\mathbb{E}[\mathbb{W}_2^2] \to 2(1-\sqrt{\alpha_n})\,\mathrm{Tr}(\bar{\Sigma}_0)$.

Unbiased accumulation keeps diversity from fully collapsing.

Theorem 1: local selection collapses diversity

With top-α selection toward a local ideal $\mathbf{u}^*$,

$$\|\bar{\mu}_t - \mathbf{u}^*\|^2 \to 0, \qquad \bar{\Sigma}_t \to \mathbf{0}.$$

Local fidelity erases global coverage.

Local selection gains fit by erasing global coverage.

Theorem 2: diversity decays by power law

If the trajectory remains in the local basin,

$$\mathrm{Tr}(\bar{\Sigma}_t) = \mathcal{O}_{a.s.}(t^{-\psi}).$$

Tail erosion has an explicit rate.

Once trapped locally, diversity shrinks at a predictable rate.

Theorem 3: drift controls true-manifold risk

For filtered distribution $\mathcal{D}_t$ and true manifold $\mathcal{D}^*$,

$$\mathcal{R}_{\mathcal{D}^*}(h_t; g^*) \le 2\ell\epsilon\,\mathbb{W}_p(\mathcal{D}_t, \mathcal{D}^*) + \mathcal{R}_{\mathcal{D}_t}(h_t; g_t) + \mathcal{O}(\ell\delta).$$

Distribution drift upper-bounds risk on true data.

For more methodology and theoretical details, please refer to the full paper.

Synthetic data selection mechanisms in low-resource communities may bias recursive loops toward model collapse.

Scan
OpenReview paper and supplement.

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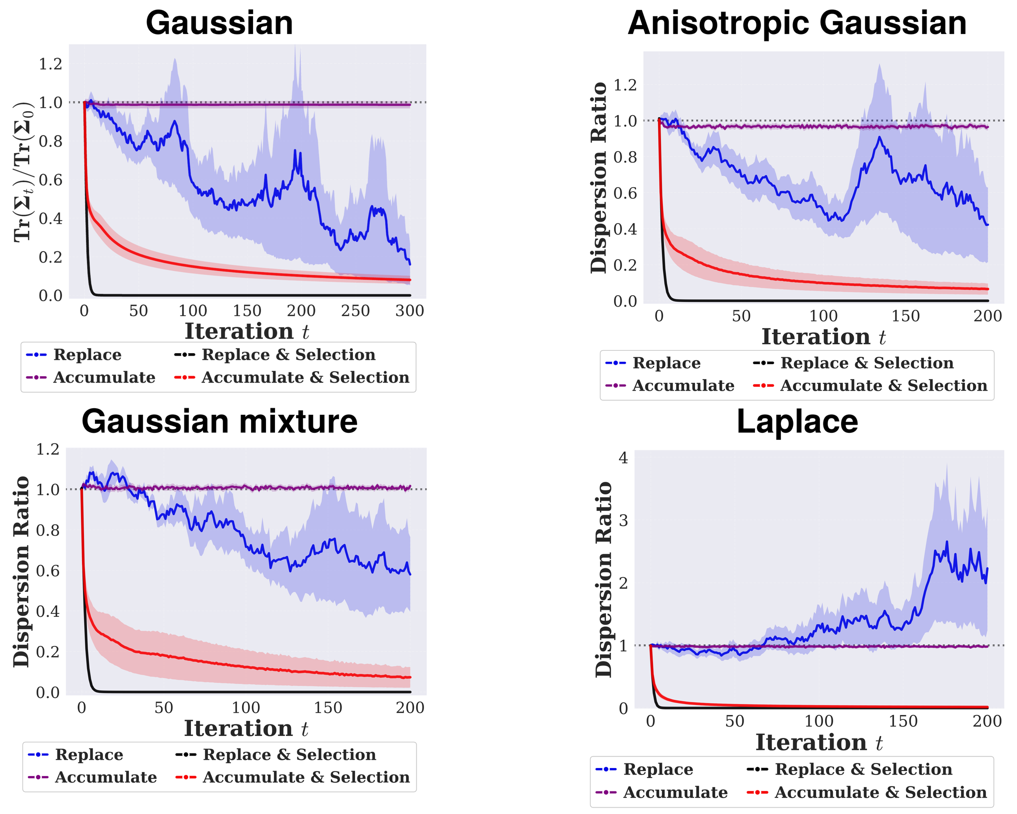
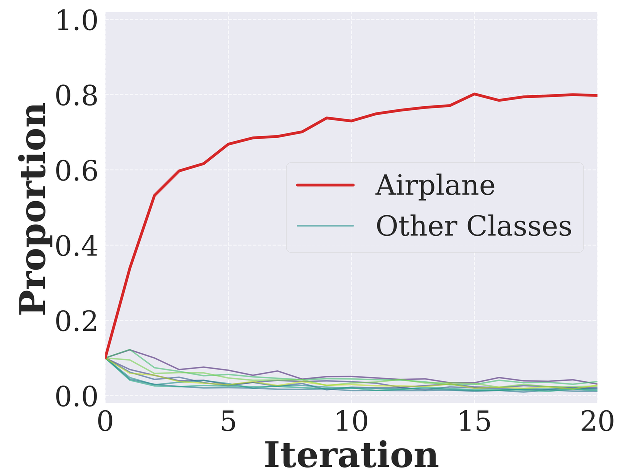
Multivariate Gaussian Modeling



Collapse Signal



Airplane-only verification moves CIFAR-10 toward Airplane and prunes other classes.

Experimental Takeaway

- Biased selection contracts synthetic distributions across modeling settings.
- Airplane-only verification concentrates CIFAR-10 on Airplane and prunes other classes.
- Unshown DDPM checks suggest non-IID references can make selection baselines lag Random, while broader references improve FID and recall.

References

[1] Shumailov et al. AI models collapse when trained on recursively generated data. Nature, 2024.
[2] Kazdan et al. Collapse or Thrive: Perils and Promises of Synthetic Data in a Self-Generating World. ICML, 2025.

For additional experimental results, please refer to the full paper.

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